Classification

Big Data y Machine Learning para Economía Aplicada

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Agenda

1. Previously...
2. Classification
3. Risk, Probability, and Classification
4. Probability Estimation
   - LPM
   - Logit
   - Probit
   - Logit EN
   - Non-Parametrics: K-Nearest Neighbors
   - Generative Models for Classification
     - Discriminant Analysis

El objetivo

- Para cualquier variable aleatoria continua y y vector de covariables W, tenemos:

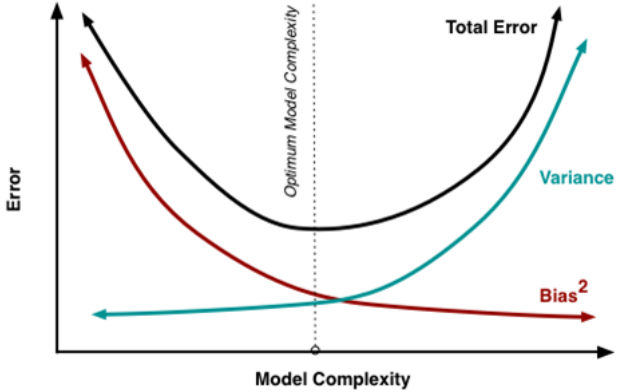
$$y = m(W) + u$$

- Bajo pérdida cuadrática, para cualquier función medible $m(W)$:

$$\hat{m} = \underset{m}{\operatorname{argmin}} \left\{ E\left[L(y, m(W)) \right] \right\}$$

Error de predicción fuera de muestra

- ML nos interesa la predicción fuera de muestra



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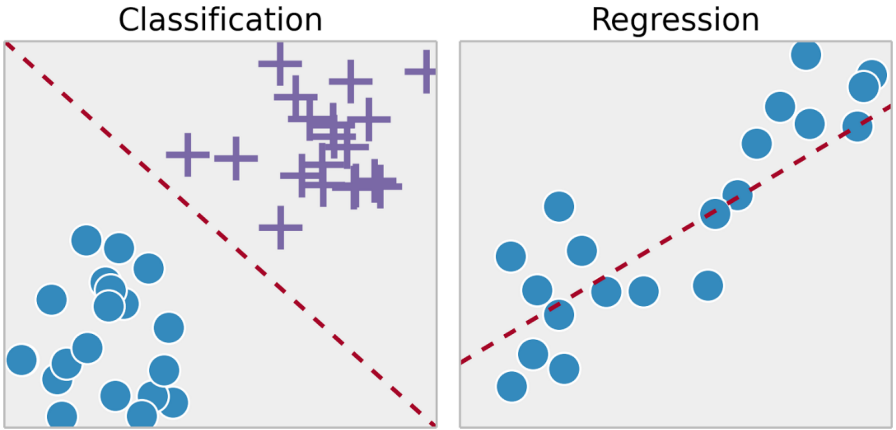
Classification



Classification: Motivation

- Many predictive questions are about classification
  - Accept someone to a graduate program or no
  - A household is bellow the poverty line
  - An individual is unemployed

Classification: Motivation

- y represents membership in a category: $y \in \{1, 2, ..., n\}$
  - Qualitative (e.g., unemployed, employed, etc)
  - Not necessarily ordered

The prediction question is, given a new X,
what is our best guess at the response category $\hat{y}$

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Risk, Probability, and Classification

- Two states of nature $Y \rightarrow i \in \{0, 1\}$
- Two actions $(\hat{Y}) \rightarrow j \in \{0, 1\}$

		$\hat{Y}$	
		0	1
Y	0	True Negative	False Positive
	1	False Negative	True Positive

Risk, Probability, and Classification

- Two actions $\hat{Y} \rightarrow j \in \{0, 1\}$
- Two states of nature $Y \rightarrow i \in \{0, 1\}$
- Probabilities
  - $p = Pr(Y = 1|X)$
  - $1 - p = Pr(Y = 0|X)$

Risk, Probability, and Classification

- Actions have costs associated to them
- Loss: $L(i,j)$, penalizes being in bin i,j
  - We define $L(i,j)$

$$L(i,j) = \begin{cases} 1 & i \neq j \\ 0 & i = j \end{cases} \tag{1}$$

- Risk: expected loss of taking action j

Bayes classifier

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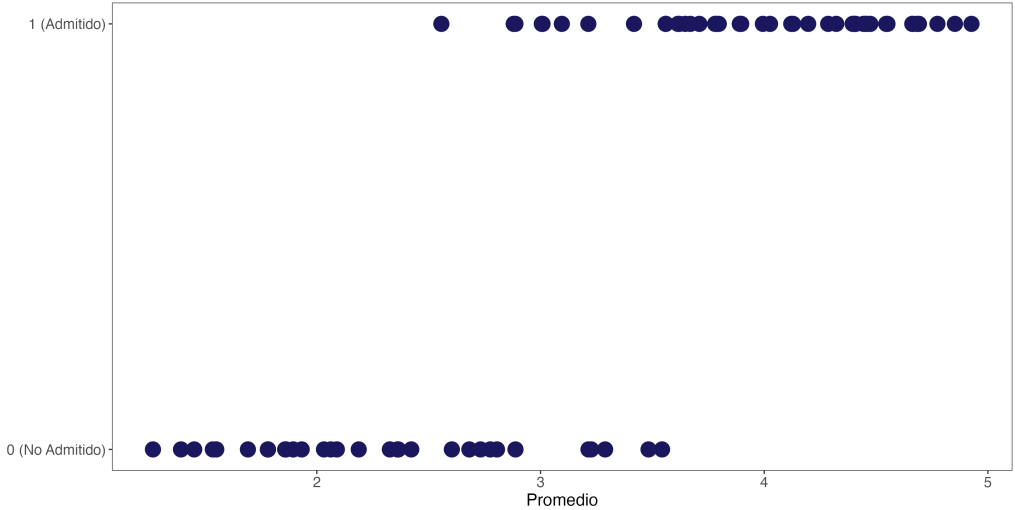
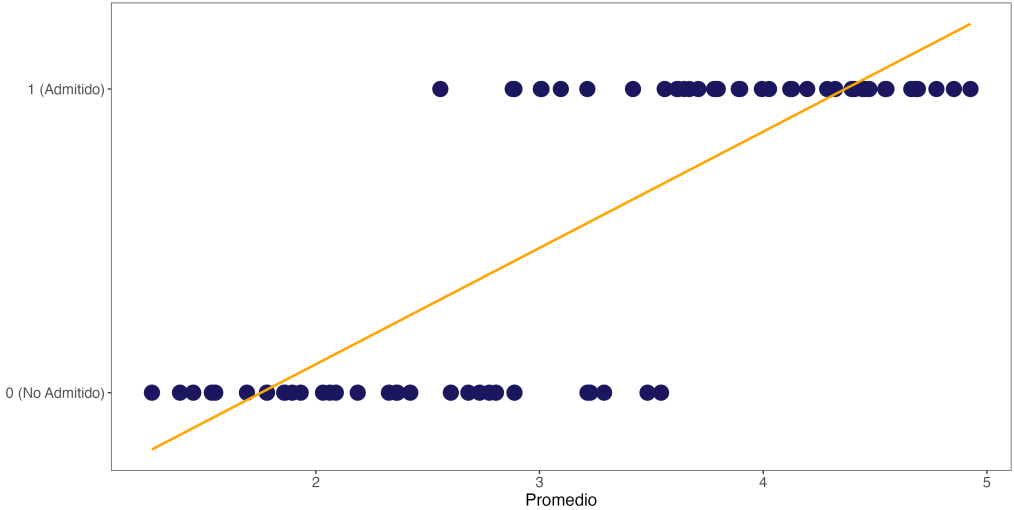
Setup

- Y is a binary random variable$\{0, 1\}$
- X is a vector of K predictors
- $p = Pr(Y = 1|X)$

$$Pr(Y = 1|X)$$



$Pr(Y = 1|X)$



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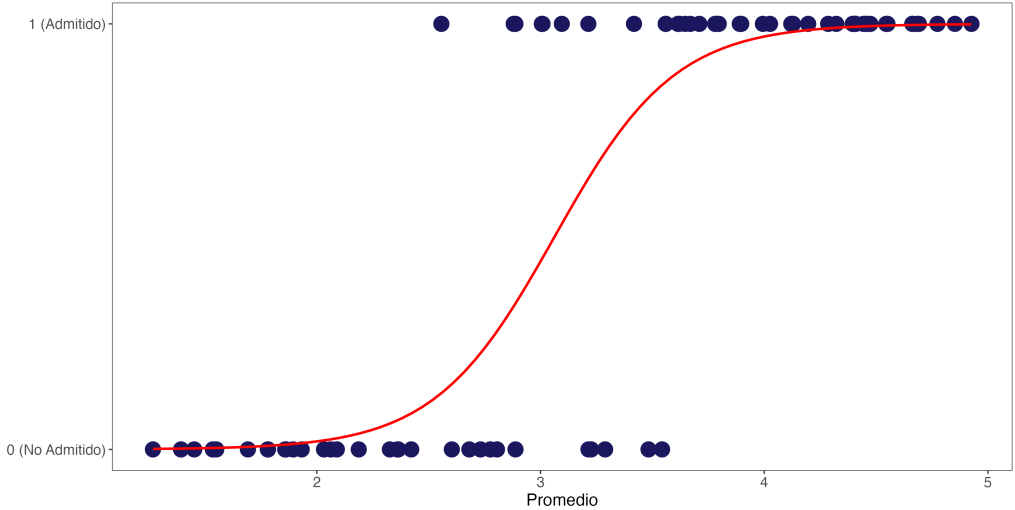


$Pr(Y = 1|X)$

Logit

- We have a sample of observations $(y_i, X_i); i = 1, \ldots, n$, where $y_i \in \{0, 1\}$ and we assume are iid

$Pr(Y = 1|X)$

Logit

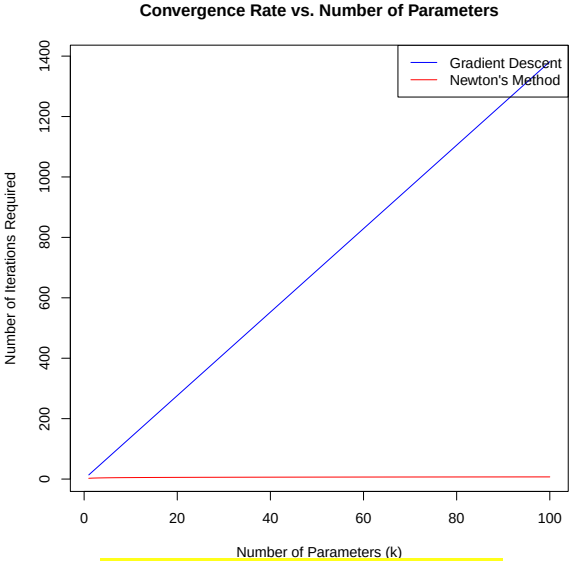
$Pr(Y = 1|X)$

Logit

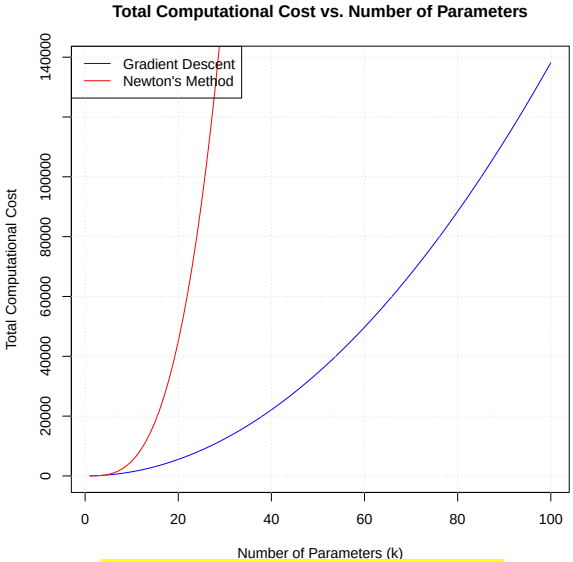
Gradient Descent vs Newton's Method

Gradient Descent vs Newton's Method

Gradient Descent vs Newton's Method



Gradient Descent vs Newton's Method



Probit

- $Pr(y = 1|X) = \Phi(X'\beta)$ where Φ is the standard normal cdf.

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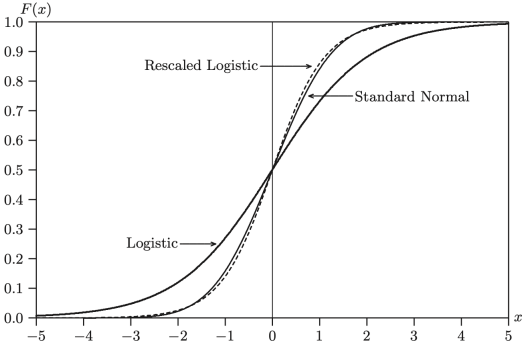


Figure 11.1 Alternative choices for $F(x)$

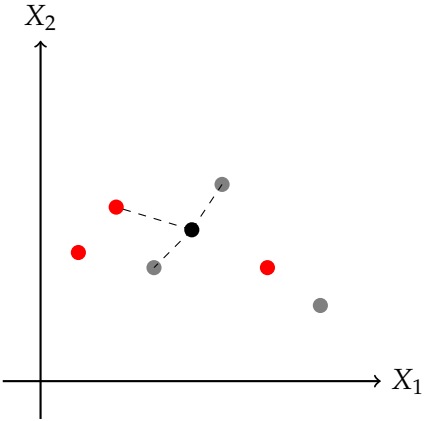
Probit

- $Pr(y = 1|X) = \Phi(X'\beta)$ where Φ is the standard normal cdf.
- In practice, the probit and logit models generally yield very similar predicted probabilities,
- There are practical reasons for favoring one or the other in some cases for mathematical convenience, in other computational convenience, but it is difficult to justify the choice of one distribution or another on theoretical grounds.

Logit EN

K-Nearest Neighbors

- K nearest neighbor (K-NN) algorithm predicts class $\hat{y}$ for x by asking *What is the most common class for observations around x?*



K-Nearest Neighbors

- K nearest neighbor (K-NN) algorithm predicts class $\hat{y}$ for x by asking *What is the most common class for observations around x?*
- Algorithm: given an input vector x_f where you would like to predict the class label
  - Find the K nearest neighbors in the dataset of labeled observations, $\{x_i, y_i\}_{i=1}^{n}$, the most common distance is the Euclidean distance:

$$d(x_i, x_f) = \sqrt{\sum_{j=1}^{p} (x_{ij} - x_{fj})^2} \tag{2}$$

  - This yields a set of the K nearest observations with labels:

$$[x_{i1}, y_{i1}], \ldots, [x_{iK}, y_{iK}] \tag{3}$$

  - The predicted class of x_f is the most common class in this set

$$\hat{y}_f = mode\{y_{i1}, \ldots, y_{iK}\} \tag{4}$$

Discriminant Analysis

$$Pr(Y = 1|X) \tag{5}$$

Linear Discriminant Analysis

Reverend Bayes to the rescue: Bayes Theorem

Discriminant Analysis

Linear Discriminant Analysis

Discriminant Analysis

Linear Discriminant Analysis

- Why is it called linear?
- Note

$$p > \frac{1}{2} \iff ln(\frac{p}{(1-p)}) > 0 \tag{6}$$

- Logit with one predictor

$$\beta_1 + \beta_2 X \tag{7}$$

- Classification: in the probability of space
- Discrimination: in the space of X
- $\beta_1 + \beta_2 X$ is the discrimination function for logit (it is lineal)

Discriminant Analysis

Linear Discriminant Analysis

- LDA?
- One predictor with $\sigma_0 = \sigma_1$ (equal variance)

Discriminant Analysis

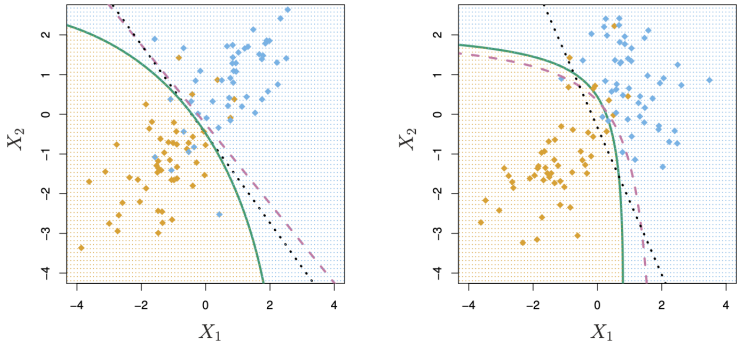
Quadratic Discriminant Analysis

- QDA assumes diferent variances for the components

Discriminant Analysis

Quadratic Discriminant Analysis

- QDA assumes diferent variances for the components



Summary

- We observe (y_i, X_i) $i = 1, \ldots, n$
- Generate probabilities
  - Logit (example)

$$p_i = \frac{e^{X_i\beta}}{1 + e^{X_i\beta}} \tag{8}$$

  - Predict

$$\hat{p}_i = \frac{e^{X_i\hat{\beta}}}{1 + e^{X_i\hat{\beta}}} \tag{9}$$

- Classification

$$\hat{y}_i = 1[\hat{p}_i > 0.5] \tag{10}$$

Accuracy

		$\hat{y}_i$	
		0	1
y_i	0	TN	FP
	1	FN	TP

- Accuracy: the fraction of predictions our model got right.

$$\frac{TP + TN}{TP + TN + FN + FP} \tag{11}$$

Example

photo from https://www.dailydot.com/parsec/batman-1966-labels-tumblr-twitter-vine/